Building Deep Learning Models with TensorFlow

Deep learning is impacting and revolutionising the tech industry. Many applications used on a day-to-day basis have been built incorporating deep learning. This article explains how the popular TensorFlow framework can be used to build a deep learning model.

Let's assume the reader has the requisite knowledge of deep learning models and algorithms. There are various frameworks that are used to build these deep learning (neural networks) models, with TensorFlow and Keras being the most popular. Figures 1 and 2 show the adoption levels and the support community for all these frameworks.

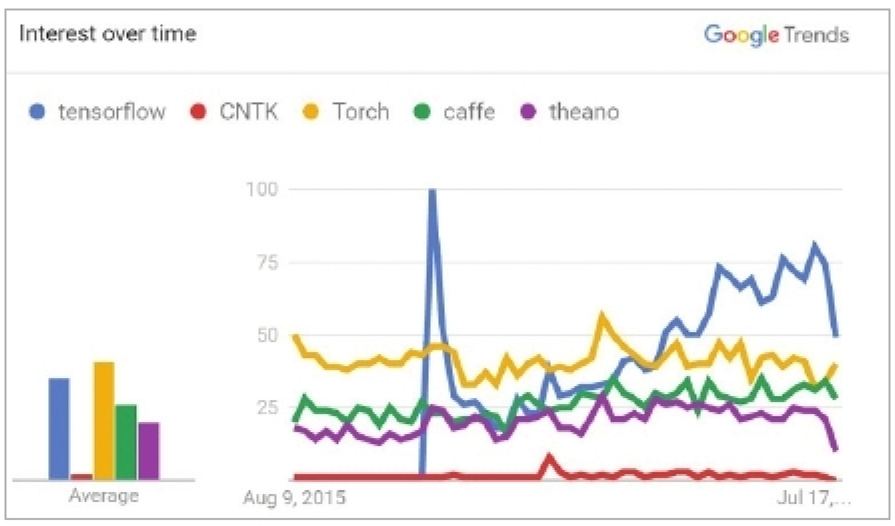

Figure 1: Interest in TensorFlow usage (Source: *https://towardsdatascience.com/what-happened-at-the-tensorflow-dev-summit-2017-part-1-3-community-applications-77fb5ce03c52)*

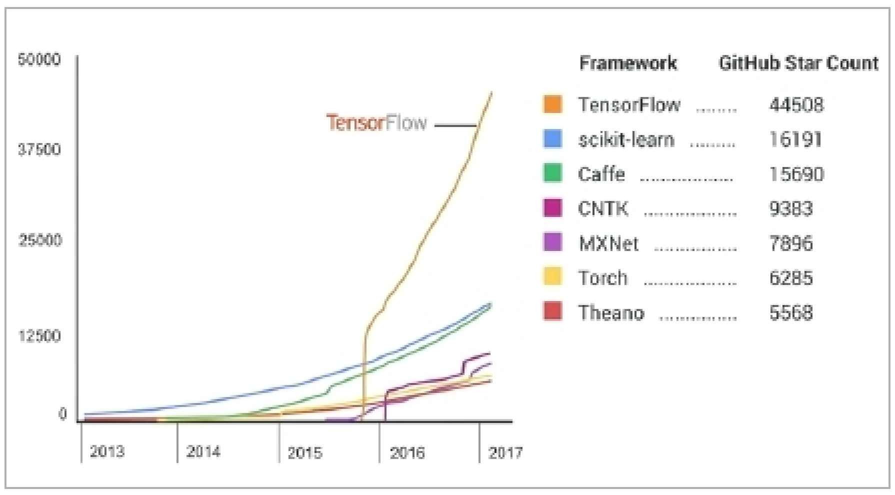

Figure 2: TensorFlow adoption (Source: *https://medium.com/@jrodthoughts/five-deep-learning-frameworks-that-you-should-know-about-and-some-data-to-back-it-up-4480d31c86a8)*

TensorFlow

TensorFlow code *depicts* 'computations' and doesn't actually perform them. To run or execute TensorFlow code, we need to create 'tf.Session' objects in Python.

Tensors: Tensors are mathematical constructs used especially in fields like physics and engineering.